Anne Vanhove

Statistical analysis of oceanographic data

— OCEA0224-A —

1) Analyze the normality of the distribution of the descriptors and propose appropriate transformations if the variables are not normally distributed.

Porosity: The Shapiro test indicates a p-value of 0.6005, therefore our data follow a normal distribution.

Grain: The Shapiro test indicates a p-value of 4.802e-06, therefore our data do not follow a normal distribution. After testing different transformations, here are the respective p-values obtained:
p.raw : 4.802206e-06
p.log : 0.001138779
p.sqrt : 6.704906e-05
p.stand : 4.802206e-06
p.stand.sqrt : 6.704906e-05
p.inverse : 0.1995378
p.inverse.sqrt : 0.03433755
The inverse transformation allows the data to be normalized.

Silt: The Shapiro test indicates a p-value of 5.385e-06, therefore our data do not follow a normal distribution. After testing different transformations, here are the respective p-values obtained:
p.raw : 5.384723e-06
p.log : 2.440522e-06
p.sqrt : 3.613907e-06
p.stand : 5.384723e-06
p.stand.sqrt : 3.613907e-06
p.inverse : 1.080506e-06
p.inverse.sqrt : 1.613590e-06
No transformation allows the data to be normalized.

$CaCO_3$: The Shapiro test indicates a p-value of 0.001633, therefore our data do not follow a normal distribution. After testing different transformations, here are the respective p-values obtained:
p.raw : 0.001632544
p.log : 0.005612565
p.sqrt : 0.004012435
p.stand : 0.001632544
p.stand.sqrt : 0.004012435
p.inverse : 0.004519594

p.inverse.sqrt : 0.005794893
No transformation allows the data to be normalized.

Shells: The Shapiro test indicates a p-value of 0.001462, therefore our data do not follow a normal distribution. After testing different transformations, here are the respective p-values obtained:
p.raw : 0.001462012
p.log : 0.001113325
p.sqrt : 0.004533987
p.stand : 0.001462012
p.stand.sqrt : 0.004533987
p.inverse : NaN
p.inverse.sqrt : NaN
No transformation allows the data to be normalized.

OrgC: The Shapiro test indicates a p-value of 0.2781, therefore our data follow a normal distribution.

TotN: The Shapiro test indicates a p-value of 0.2924, therefore our data follow a normal distribution.

CN: The Shapiro test indicates a p-value of 0.001941, therefore our data do not follow a normal distribution. After testing different transformations, here are the respective p-values obtained:
p.raw : 0.001940993
p.log : 0.02612359
p.sqrt : 0.008279552
p.stand : 0.001940993
p.stand.sqrt : 0.008279552
p.inverse : 0.265171
p.inverse.sqrt : 0.1048622
The inverse function is the transformation that allows the data to follow a normal distribution.

TotFe: The Shapiro test indicates a p-value of 0.0312, therefore our data do not follow a normal distribution. After testing different transformations, here are the respective p-values obtained:
p.raw : 0.001940
p.log : 0.026123
p.sqrt : 0.008279
p.stand : 0.001940
p.stand.sqrt : 0.008279
p.inverse : 0.265171
p.inverse.sqrt : 0.104862
The inverse function is the transformation that allows the data to follow a normal distribution.

TotMn: The Shapiro test indicates a p-value of 0.0009631, therefore our data do not follow a normal distribution. After testing different transformations, here are the respective p-values obtained:
p.raw : 0.0009631101
p.log : 0.03981542
p.sqrt : 0.08309115

p.stand : 0.0009631101
p.stand.sqrt : 0.08309115
p.inverse : 0.06778187
p.inverse.sqrt : 0.4932306
The data follow a normal distribution when transformed using the inverse function or the inverse function followed by a square root.

TotP: The Shapiro test indicates a p-value of 0.9189, therefore our data follow a normal distribution.

Regarding species, they also did not follow a normal distribution, and it will therefore be necessary to apply a Hellinger transformation before using them in further analyses.

2) Calculate biodiversity indices and briefly map their spatial distribution: use longitude and latitude as x and y, and vary point size and intensity according to the values.

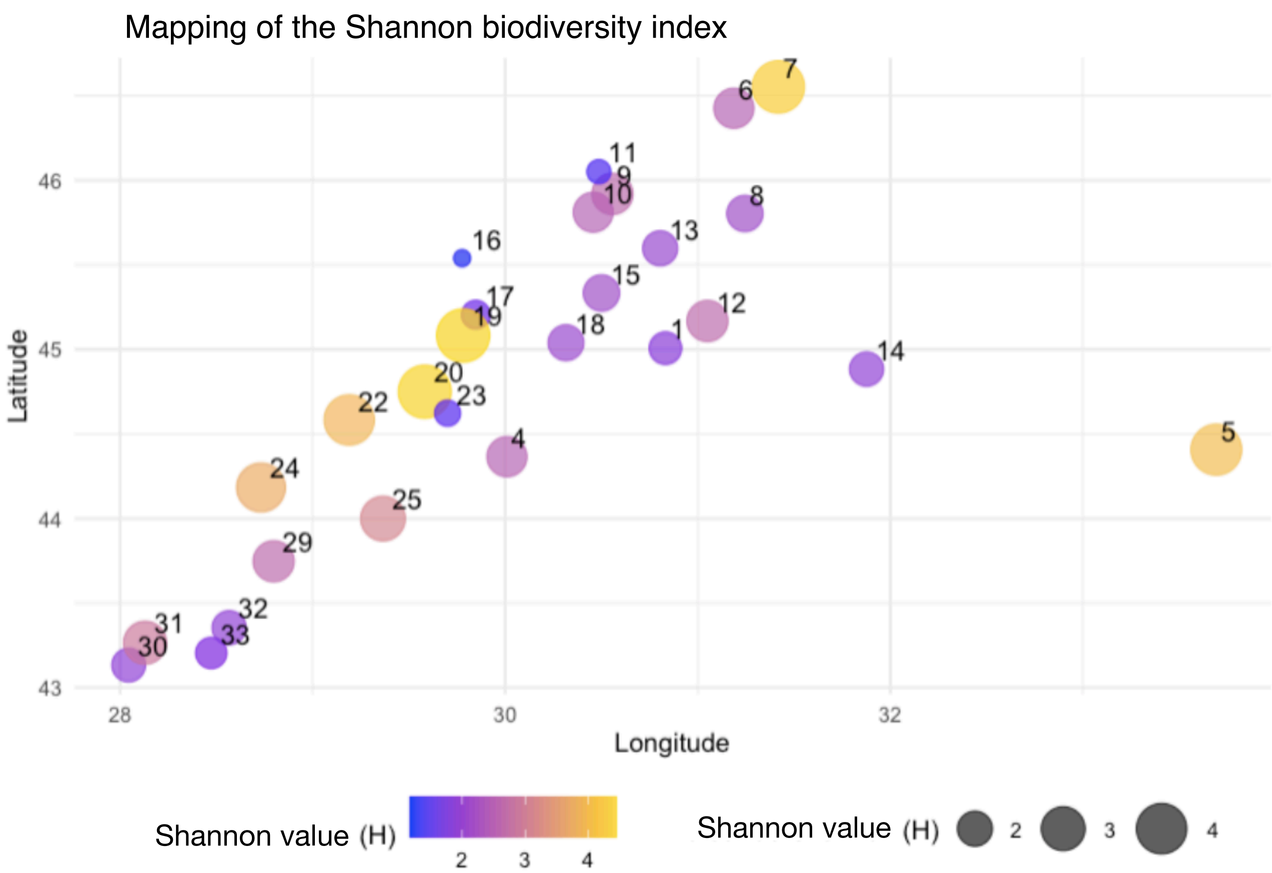


Figure 1: Map of a biodiversity indicator (Simpson index value) across the different sites.

Thanks to this map, we observe that sites 7, 19, and 20 have high Simpson values (around 4 or higher), indicating greater biodiversity than in the other sites. Conversely, sites 11, 16, and 17 show the lowest values (around 1), indicating lower biodiversity.

3) Calculate distance matrices (compare several appropriate metrics) between sites based on species distribution and environmental variables. Interpret.

Distance matrices based on species distribution and environmental variables indicate similar sites in magenta and dissimilar sites in blue. Several metrics were tested, namely Bray-Curtis (raw data), Bray-Curtis (log-transformed), Chord, and Hellinger (**Appendix 1a**). For the analysis of site matrices based on environmental variables and species, a Q-mode approach was used.

The most appropriate metric for studying species distribution is the Hellinger transformation, as it minimizes the impact of negative values (**Figure 2**). Given that our dataset contains many zeros, it is important that these zeros are not interpreted as similarities between sites.
Thus, we observe that sites 21, 1, 18, 12, and 25 are similar. The same applies to sites 14 and 4, and to sites 6, 24, 20, 19, and 22. The remaining sites, shown in blue, do not appear to share the same species.

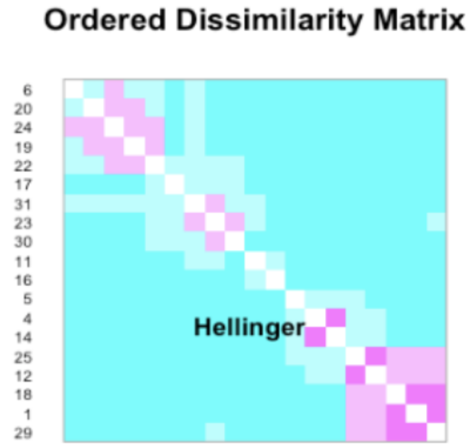


Figure 2: Hellinger distance matrix representing the biodiversity per site.

Regarding the study of environmental variables, I used the Chord matrix, as this method is insensitive to scale differences and is therefore suitable for analyzing environmental variables with different units (**Figure 3**; **Appendix 1b**). The interpretation method is the same as for the biodiversity matrix. Thus, we observe that sites 24 and 25 are weakly correlated, and that site 30 is not similar to any other site. Two groups emerge from this matrix: one composed of sites 18, 1, 4, 14, 23, and 29, and the other of sites 22, 17, 20, 31, 19, 16, 6, and 11. These two groups consist of sites that are very similar to each other (magenta color).

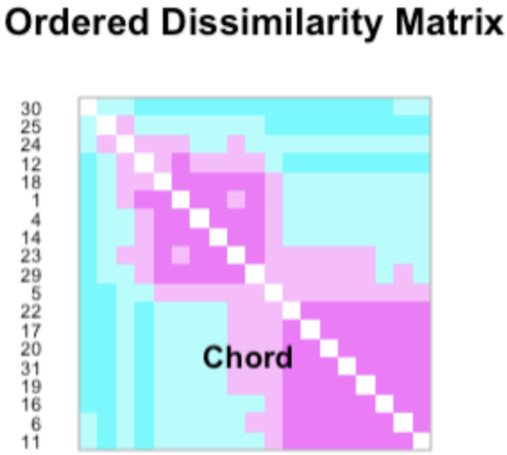


Figure 3: Chord distance matrix with environmental variables per site.

4) Perform a clustering of sites based on species and environmental variables. Choose the most appropriate method and justify it. Compare.

<u>*Clustering of sites based on species:*</u>

First, I calculated the different cophenetic correlations using various methods: the "Single" method, the "Complete" method, the "Average" method, the "Centroid" method, and the "Ward" method. It can be observed that the same dendrograms are obtained with the Complete, Average, and Centroid methods (**Figure 4**). The highest correlation is obtained with the "Average" method (0.958). By plotting the fusion levels of the nodes in the dendrogram, a significant jump is observed between levels 5 and 6 (**Appendix 2**). I therefore chose k = 5 and obtained a contingency table that showed a perfect correspondence between the "Complete" and "Average" methods, meaning that both methods form clusters composed of the same sites (**Figure 5**). The "Average" method was used for the rest of the analysis.

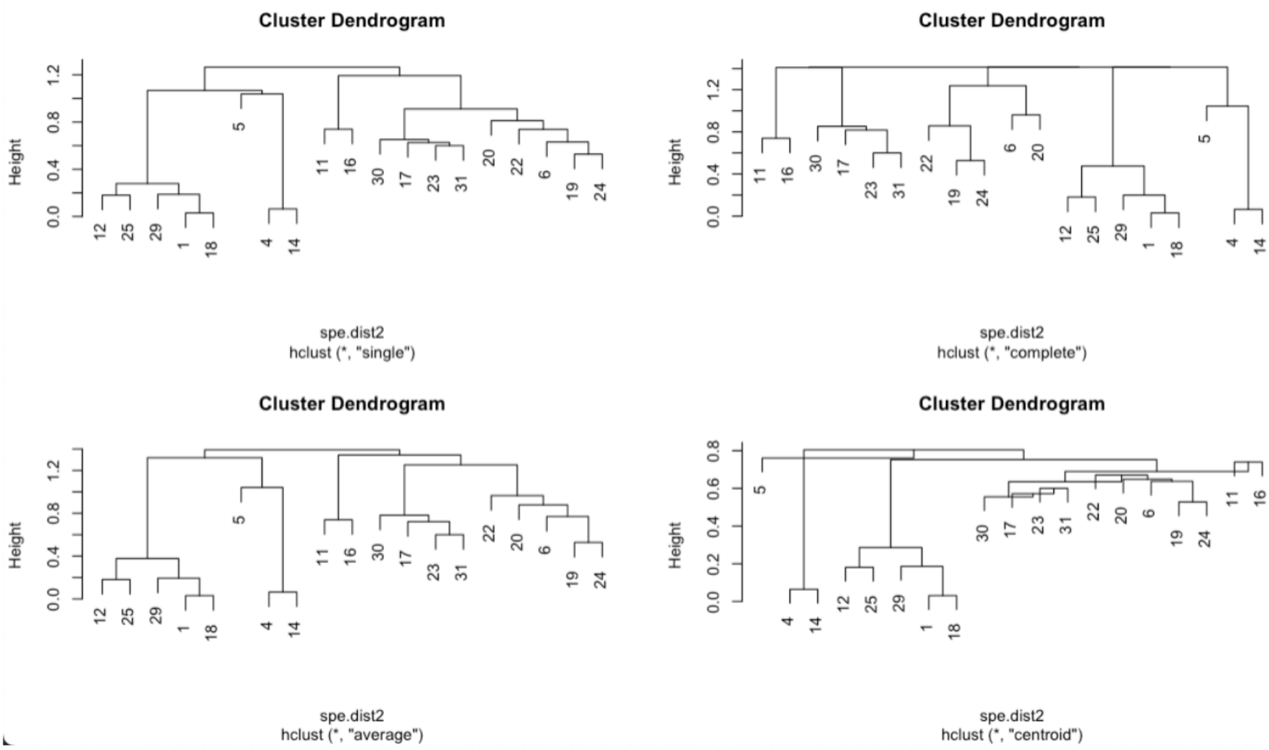


Figure 4: Dendrograms showing the relationships between sites based on their respective biodiversity. The first dendrogram uses the "Single" method, the second (on the right) uses the "Complete" method, the one at the bottom left was generated using the "Average" method, and the one at the bottom right using the "Centroid" method.

```
> table(spebc.average.g2, spebc.complete.g2)
                spebc.complete.g2
spebc.average.g2 1 2 3 4 5
               1 5 0 0 0 0
               2 0 3 0 0 0
               3 0 0 5 0 0
               4 0 0 0 2 0
               5 0 0 0 0 4
```

Figure 5: Contingency table of the "Complete" and "Average" methods for the species dendrogram.

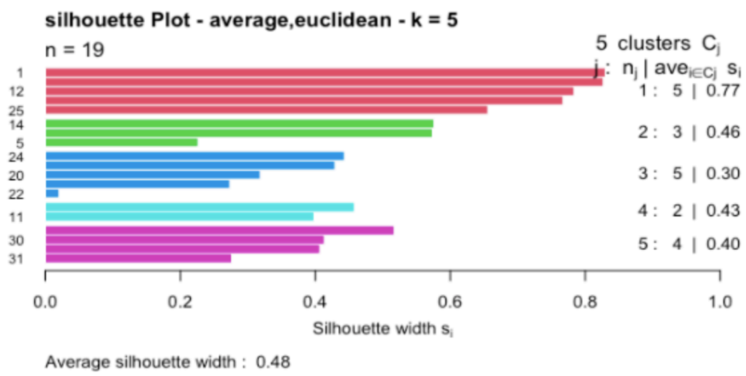


Figure 6: Silhouette plot of each cluster from the dendrogram for the biodiversity analysis.

Although the silhouette method indicated an optimal number of clusters equal to 6, I decided to retain 5 clusters because the 6th had a silhouette value of 0, indicating poor clustering (**Figure 6**). The resulting clusters have silhouette values of 0.77 for cluster 1, 0.46 for cluster 2, 0.3 for cluster 3, 0.43 for cluster 5, and 0.40 for the last cluster.

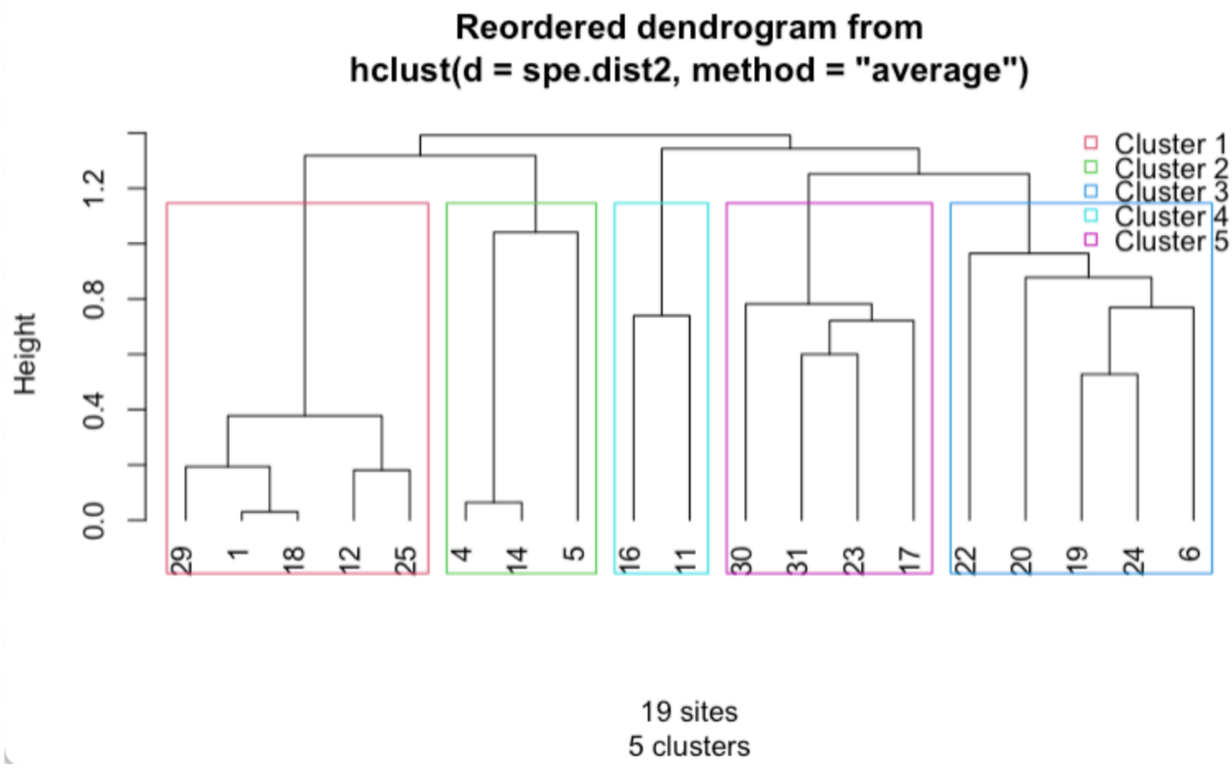


Figure 7: Final dendrogram showing the clusters of the different sites based on their biodiversity.

In order to observe the distribution of different species within clusters, as well as their contribution to the formation of each cluster, a facet wrap was produced (Figure 8).

From these graphs, it appears that few species are abundant, except for *Mytilus galloprovincialis*, *Modiolus phaseolinus*, *Melinna palmata*, and *Nereis diversicolor*. Stations belonging to cluster 1 seem to host all the studied species, with relatively high silhouette values for most of them, around 0.7. However, *Mytilus galloprovincialis* (high abundance), *Modiolus phaseolinus*, and *Terebellides stroemi* (very low abundance) are the most represented species in the sites of cluster 1. Sites in cluster 2 appear to share the presence of *Modiolus phaseolinus*, *Terebellides stroemi,* and *Molluga euprocta*. As for cluster 3, its sites seem to be characterized by *Mya arenaria*, *Nereis diversicolor*, *Nereis rava*, and *Nereis succinea*. Cluster 4 includes sites sharing the presence of *Melinna palmata*, *Pectinaria koreni*, *Nemertini varia*, *Nereis rava*, and *Nereis succinea*. Finally, the last cluster is characterized by the presence of *Melinna palmata*, *Nassa reticulata*, and *Nereis diversicolor*.

Thus, through these graphs, we observe that each cluster is characterized by species that are rarely or not found in the other clusters.

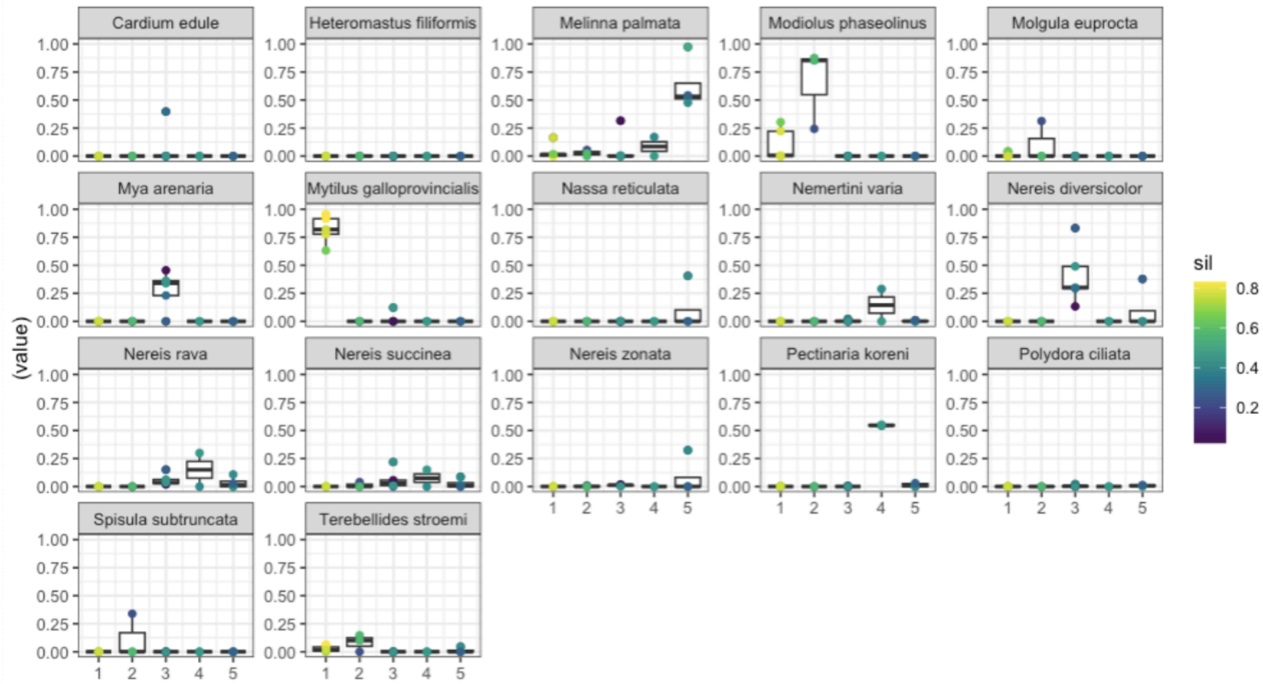


Figure 8: Graphs showing the relative abundance of each species across the different clusters (x-axis), along with their associated silhouette values (sil). Silhouette values range from 0 (blue) to 1 (yellow).

Clustering of sites based on environmental variables:

The same methodology as for clustering sites based on biodiversity was used to study environmental variables. The dendrograms obtained using the different methods are shown in

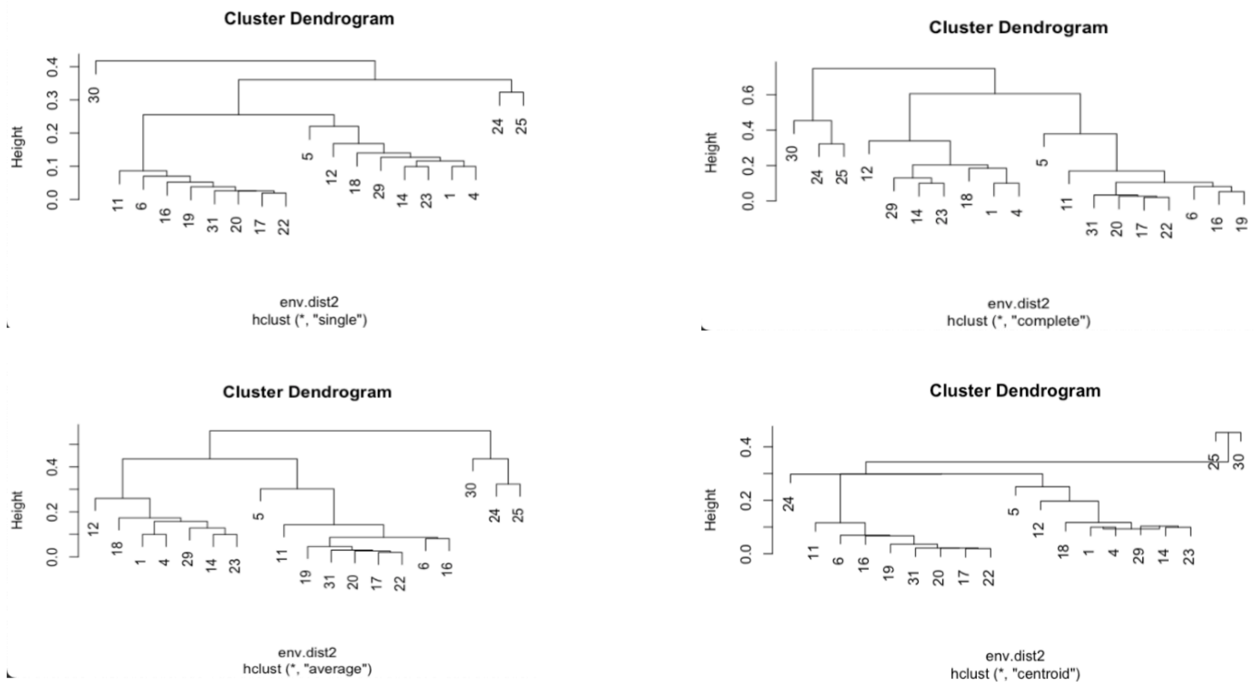


Figure 9: Dendrograms showing the relationships between sites based on their respective biodiversity. The first dendrogram uses the "Single" method, the second (on the right) uses the "Complete" method, the one at the bottom left was generated using the "Average" method, and the one at the bottom right using the "Centroid" method.

Figure 9, where it can be observed that the "*Average*" and "*Complete*" methods produce the same dendrograms.

Thus, the highest correlation is obtained with the "Average" method (0.906). By plotting the fusion levels of the nodes in the dendrogram, a significant jump is observed between levels 2 and 3, as well as between levels 4 and 5 (*Appendix 3*). Different values of k were tested, and the most relevant results corresponded to k = 4. Indeed, a contingency table was obtained and showed a perfect correspondence between the "Complete" and "Average" methods, meaning that both methods form the same clusters (*Figure 10*). Therefore, the analysis was continued using the "Average" method. In addition, the silhouette method indicated an optimal number of clusters equal to k = 4, which is also the value for which the highest silhouette scores were obtained (*Figure 11*).

```
> table(envbc.average.g2, envbc.complete.g2)
                 envbc.complete.g2
envbc.average.g2 1 2 3 4
               1 7 0 0 0
               2 0 9 0 0
               3 0 0 2 0
               4 0 0 0 1
```

Figure 10: Contingency table of the "Complete" and "Average" methods for the environment dendrogram.

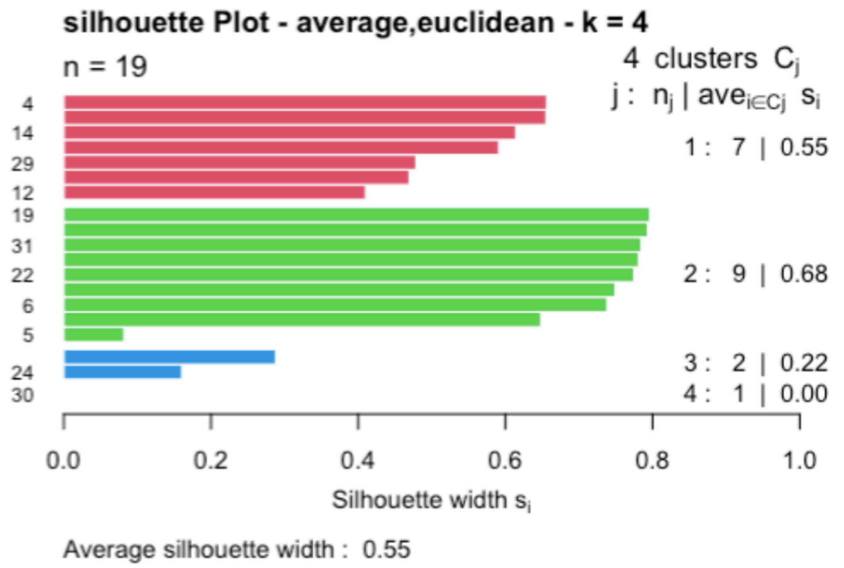


Figure 11: Silhouette plot of each cluster from the dendrogram for the analysis of environmental variables.

However, regardless of the value of k used, at least one cluster consistently had a silhouette value equal to 0. Thus, the first cluster includes sites 29, 23, 14, 18, 4, 1, and 12. In the environmental distance matrices, these sites were identified as highly similar. The second cluster includes sites 11, 6, 31, 17, 22, 19, 20, 16, and 5, and has the highest silhouette value (0.88); these groups were also clustered together in the similarity matrix. Finally, cluster 3 groups sites 24 and 25, and cluster 4 contains only site 30 (**Figure 12**), the latter being dissimilar to all other sites in the dissimilarity matrix (**Figure 3**).

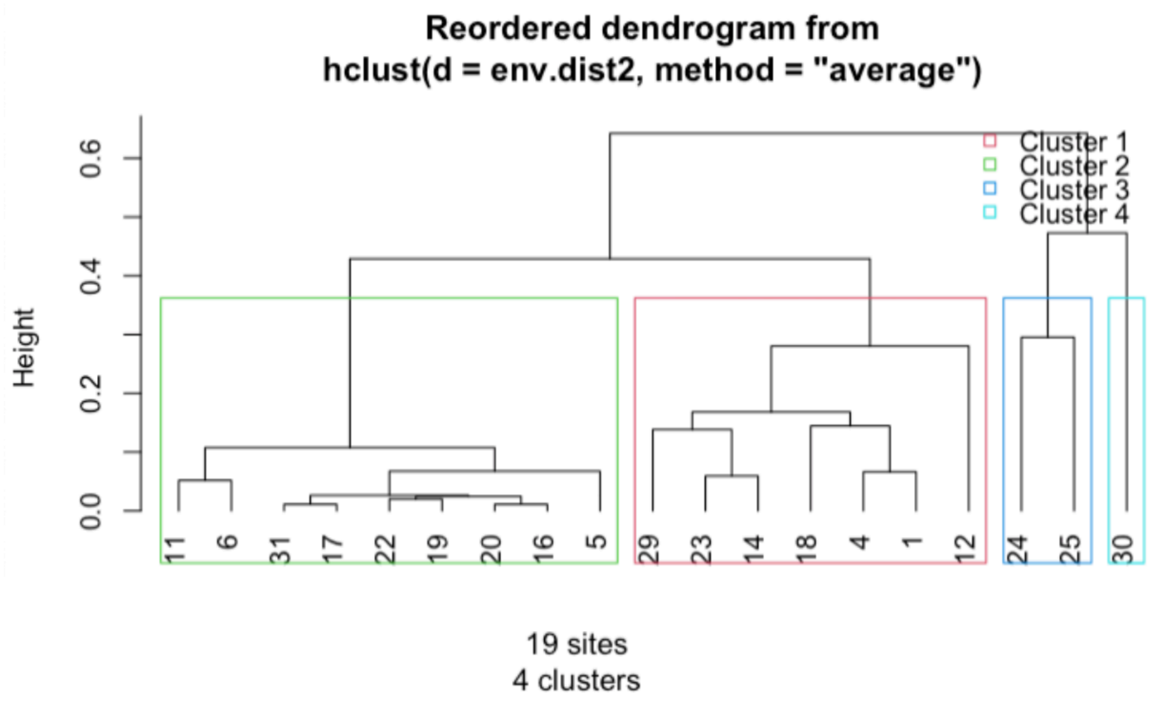


Figure 12: Final dendrogram showing the clusters of the different sites based on environmental conditions.

In order to observe the distribution of the different environmental variables within the clusters, as well as their contribution to the formation of each cluster, a facet wrap was performed (**Figure 13**).

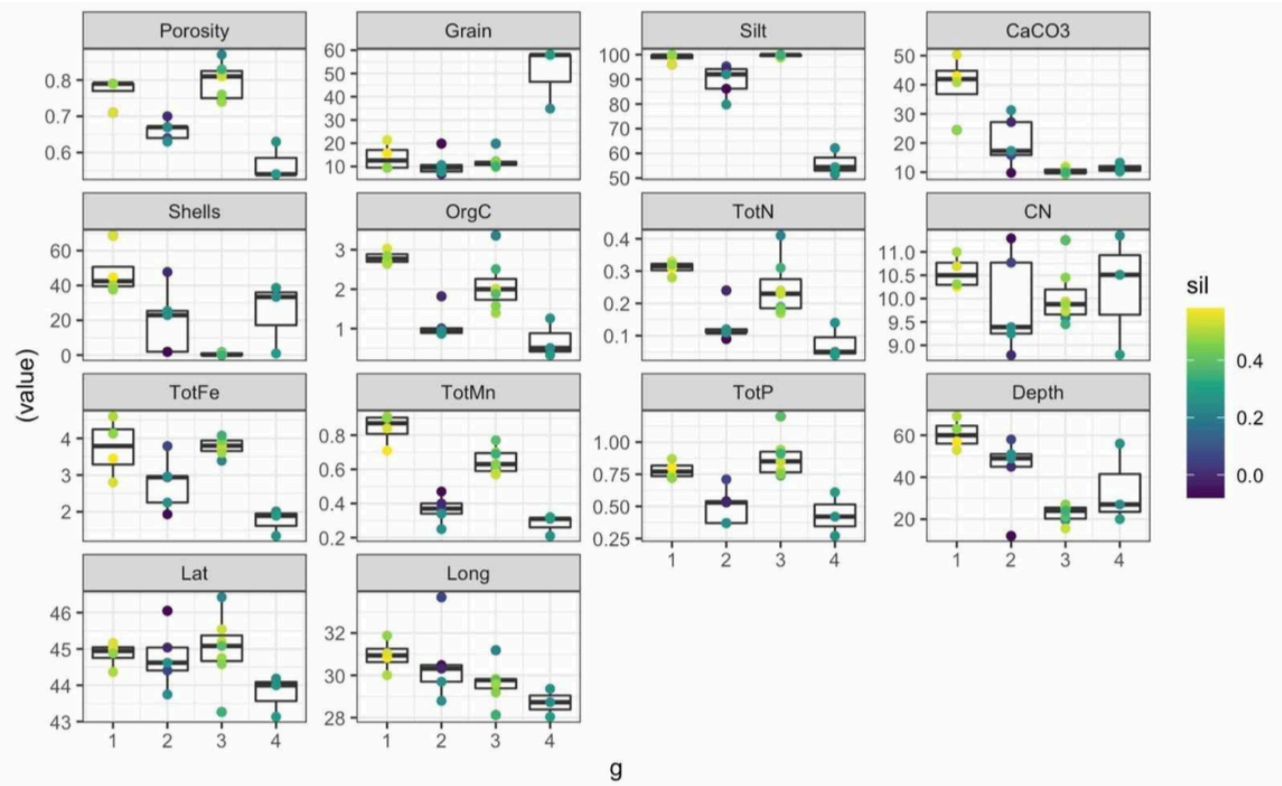


Figure 13: Graphs showing the relative abundance of each species across the different clusters (x-axis), along with their associated silhouette values (sil). Silhouette values range from 0 (blue) to 1 (yellow).

Thanks to these graphs, we observe that stations in cluster 1 have soils with high porosity, greater depths, low grain size, and high amounts of shells and $CaCO_3$. Cluster 2 shows similar trends to cluster 1 but with lower levels. Cluster 3 includes sites with high porosity, silty substrates, low $CaCO_3$ and shell content, and relatively high nutrient levels. Finally, cluster 4 consists of sites with sandy substrates, low $CaCO_3$, and low nutrient content.

5) Perform a PCA analysis of both matrices. Compare the two ordinations and interpret the role of environmental variables in species distribution.

The use of PCA makes it possible to reduce the number of dimensions while retaining a maximum amount of information (variance). The principal axes of a PCA help visualize the variables that contribute most to differences in the data. In addition, PCA highlights the relationships within the data: positive correlation, negative correlation, or absence of correlation. Thus, the data are represented in a 2D plot, making them easier to interpret. Regarding the orientation of the arrows, two perpendicular arrows indicate no correlation between two variables. Conversely, two parallel arrows indicate a positive correlation, while two arrows pointing in opposite directions (180°) represent negatively correlated variables.

<u>*PCA of the environmental variables matrix:*</u>

By examining the cumulative values, we find that approximately 73% of the variance is explained by the first two principal components. Therefore, it is appropriate to use these two components for the rest of the analysis. Two types of scaling can be observed: scaling 1 (on the left), generally used for relationships between individuals, and scaling 2, which is used to understand the influence of variables, where arrow length is proportional to the contribution of the associated variable. Thus, the analysis was based on scaling 2.

PC1 (horizontal axis) is dominated by sediment grain size, while PC2 (vertical axis) is dominated by depth and carbonate chemistry (**Figure 14**). Stations located to the right of the

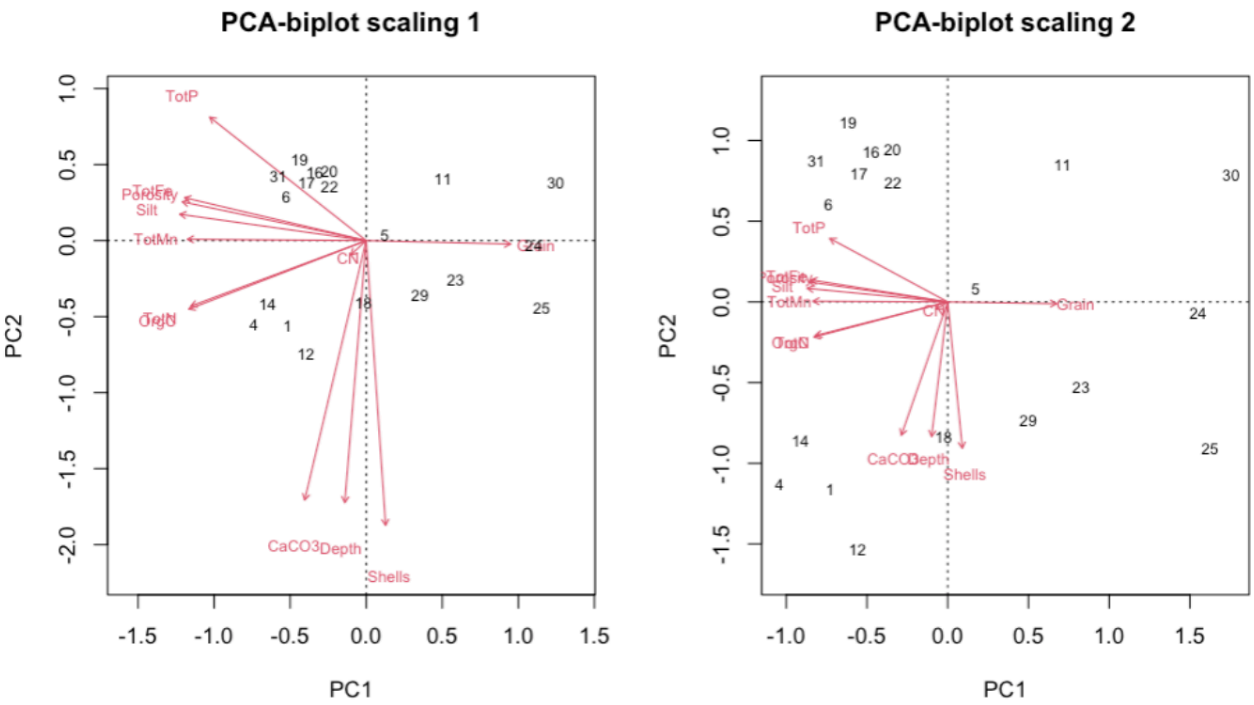


Figure 14: PCA of the environmental variables matrix with site representation.

vertical axis, between 0 and 1.5, likely have sandy substrates, whereas stations on the left, between 0 and -1, likely have muddier substrates.

Sites 6, 16, 17, 19, 20, 22, and 30 all have relatively shallow bathymetry, between 15.6 and 27 meters, and probably an intermediate substrate, neither muddy nor sandy. It is also observed that depth, $CaCO_3$, and shell content are strongly correlated, suggesting that site 18 may host a higher abundance of calcareous organisms. Finally, the variables Silt, TotP, TotN, Porosity, TotMn, and TotFe are negatively correlated with grain size. They appear to be associated with muddy substrates rich in organic matter and nutrients. Thus, sites 1, 4, 12, and 14 likely have this type of substrate and may therefore host different species compared to sites 23, 24, and 25, which have sandier substrates.

<u>*PCA of the species matrix:*</u>

First, the eigenvalues of the first three principal components were respectively 23%, 15%, and 12%. By examining their cumulative values, we find that approximately 70% of the variance is explained by the first three principal components. Scaling 3 is not presented in this work as it provided the same information as scaling 2 shown below (**Figure 15**).

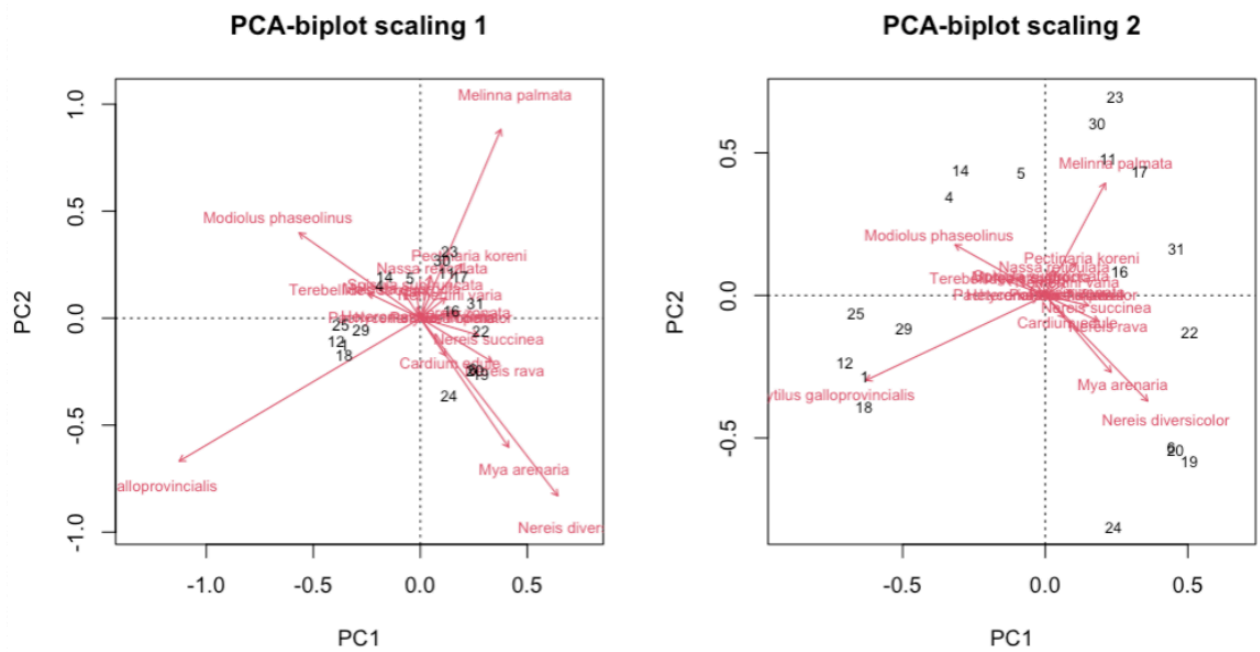


Figure 15: PCA of the species matrix with site representation.

Two types of scaling can be observed: scaling 1 (on the left) is generally used for relationships between individuals, while scaling 2 is used to understand the influence of variables, with arrow length proportional to the contribution of the associated variable. Thus, the analysis was based on scaling 2 for interpretation.

In this PCA, it can be observed that the species *Nereis diversicolor* and *Mya arenaria*, an annelid and a bivalve, are strongly correlated and are therefore often found together in the same sites. They also appear to be particularly abundant at sites 6, 20, and 19.

These two species are negatively correlated with *Modiolus phaseolinus*, another bivalve, which means they are rarely found in the same environment. As for *Mytilus galloprovincialis*, the Mediterranean mussel, it is associated with sites 1, 12, 18, and 29, and appears to be negatively correlated with *Melinna palmata*, an annelid, which is linked to sites 11, 17, 30, and 23. In addition, *M. galloprovincialis* shows no particular correlation with *N. diversicolor* and *M. arenaria*, nor with *M. phaseolinus*. Most of the other species are plotted very close to the axis, indicating that they have little influence in this PCA on the distribution of sites. However, since the PCA only represents 51% of the total variance, it is possible that these species would appear more influential when considering additional axes. It is also possible that their arrows are short due to their low abundance. Overall, this PCA suggests that the dominant species are *M. galloprovincialis*, *N. diversicolor*, *M. arenaria*, *M. phaseolinus*, and *M. palmata*.

6) Interpret the results based on the functioning of the system.

The study was conducted in the Black Sea near the Danube. Thus, habitats can be divided into three groups: the Danube mouth, sites north of the Danube, and sites to the south. Sites located near the river mouth (16, 17, 19, 20, and 22) formed cluster 2 and showed high nutrient input, silty substrates, and biodiversity mainly composed of detritivores such as *Nereideae*. These sites had the highest biodiversity indices in the study (**Figure 1**) and are located along the coast with relatively shallow bathymetry near the Danube mouth.

Sites located north of the study area (1, 8, 9, 10, 13, 14, 15, and 18) are influenced by nutrient and organic matter inputs due to the presence of a gyre transporting Danube outflow northward. These sites belong to environmental cluster 1, characterized by high organic matter content and silty sediments. As a result of these nutrient inputs, filter-feeding organisms such as *Modiolus phaseolinus* and *Mytilus galloprovincialis* are found there, as shown by the PCA (Figure 15).

Sites located south of the study area (4, 22, 24, 25, and 29) are more sheltered from Danube outflows. These sites are included in environmental cluster 3. They are characterized by lower sedimentation rates, lower organic matter input, sandy substrates, and higher grain size. These sites mainly host suspension feeders, as indicated by the PCA, including *Mytilus galloprovincialis* and *Cardium edule*.

From a broader perspective, sites located on the continental shelf (1, 4, 12, 14, and 18) have environments rich in organic matter, high carbonate content, and are covered with silt. They host bivalves such as *Modiolus phaseolinus* and *Mytilus galloprovincialis*, which are also found in northern sites influenced by the gyre and receiving significant organic matter input.

Overall, the presence of the Danube mouth strongly influences the substrate found on the seafloor. The closer a site is to the river mouth, the more silty the substrate becomes. Conversely, as distance from the mouth increases, the substrate becomes sandier, with increasing grain size. Moreover, the Danube represents a major source of organic matter and nutrients, leading to high diversity of deposit feeders and especially filter feeders in nearby areas.The presence of the gyre enables the transport of organic matter and nutrients over longer distances, explaining similarities between northern sites and those near the river mouth.Thus, the influence of the river can extend far beyond the immediate vicinity of the estuary and significantly shape the composition of benthic communities in surrounding areas.

7) Perform a PCA analysis of the sites based on biodiversity indices.

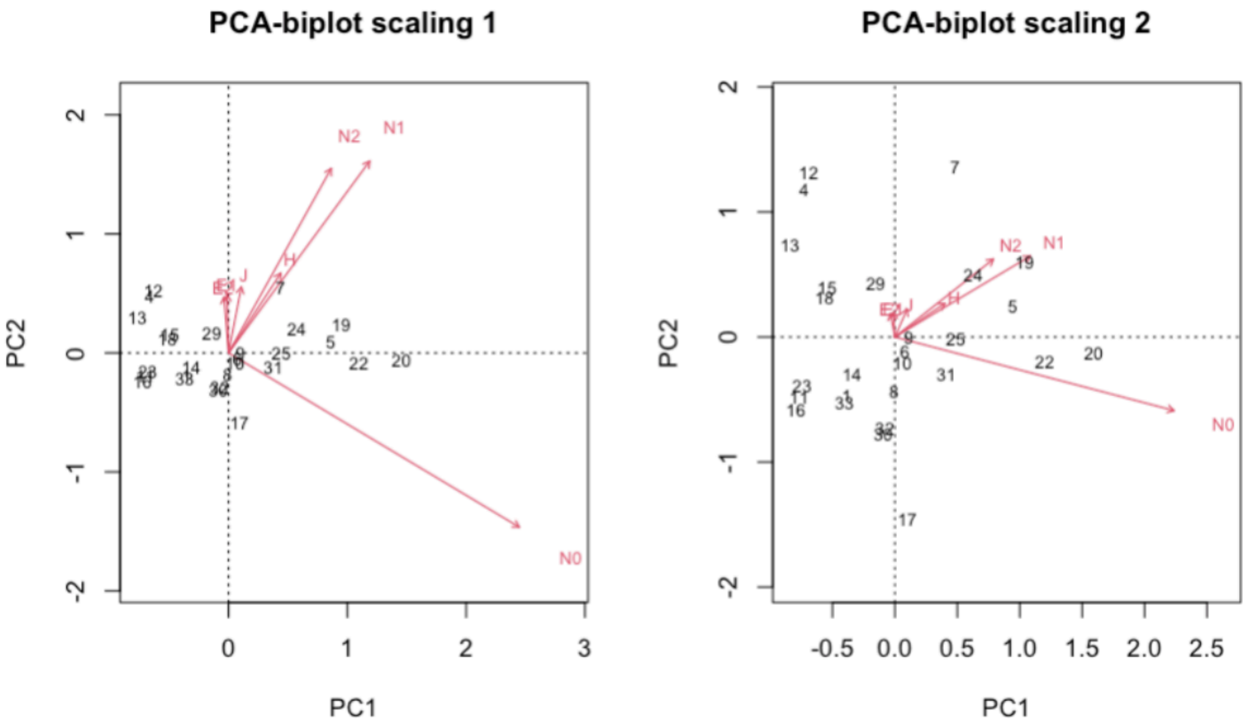
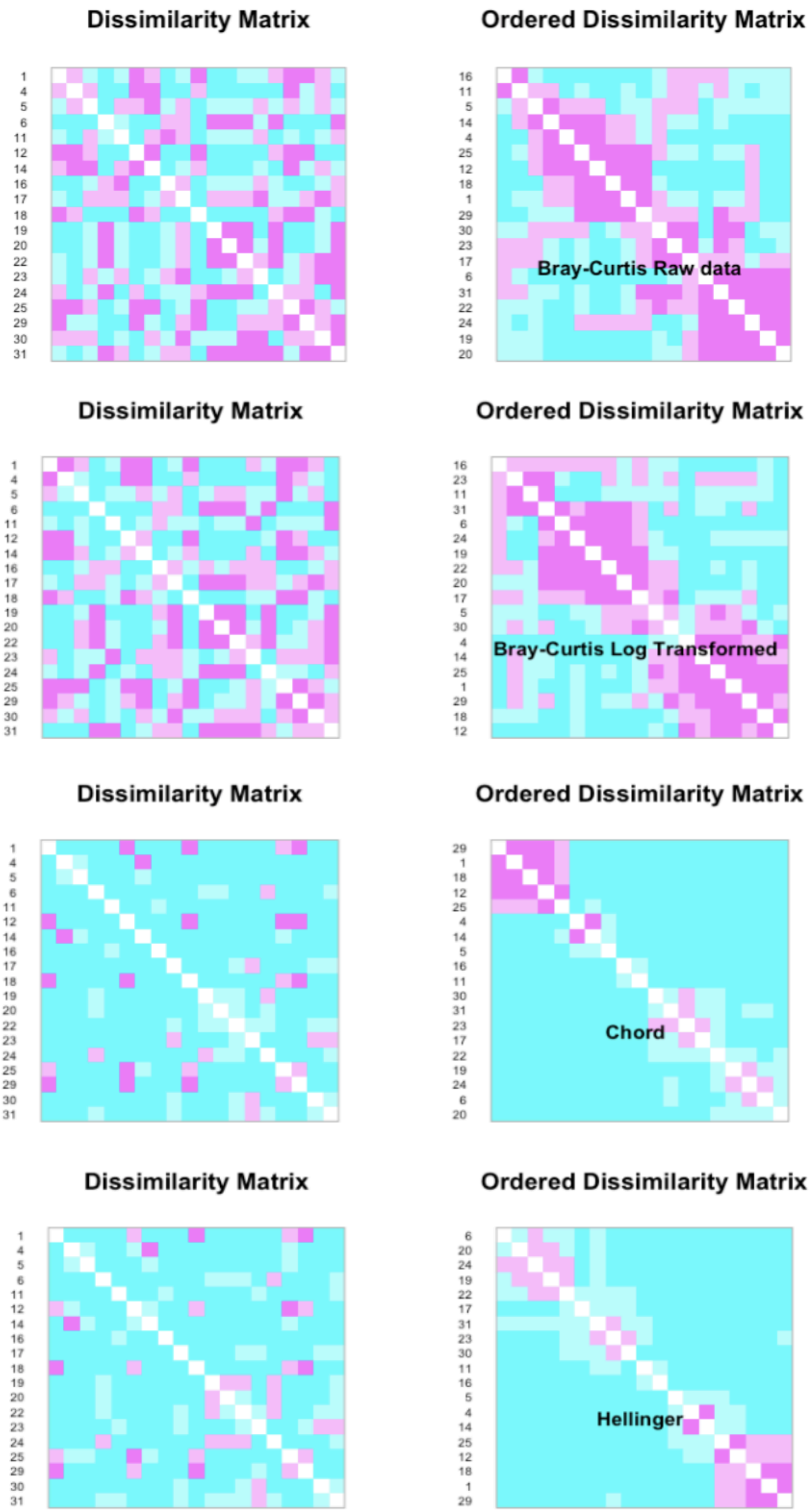


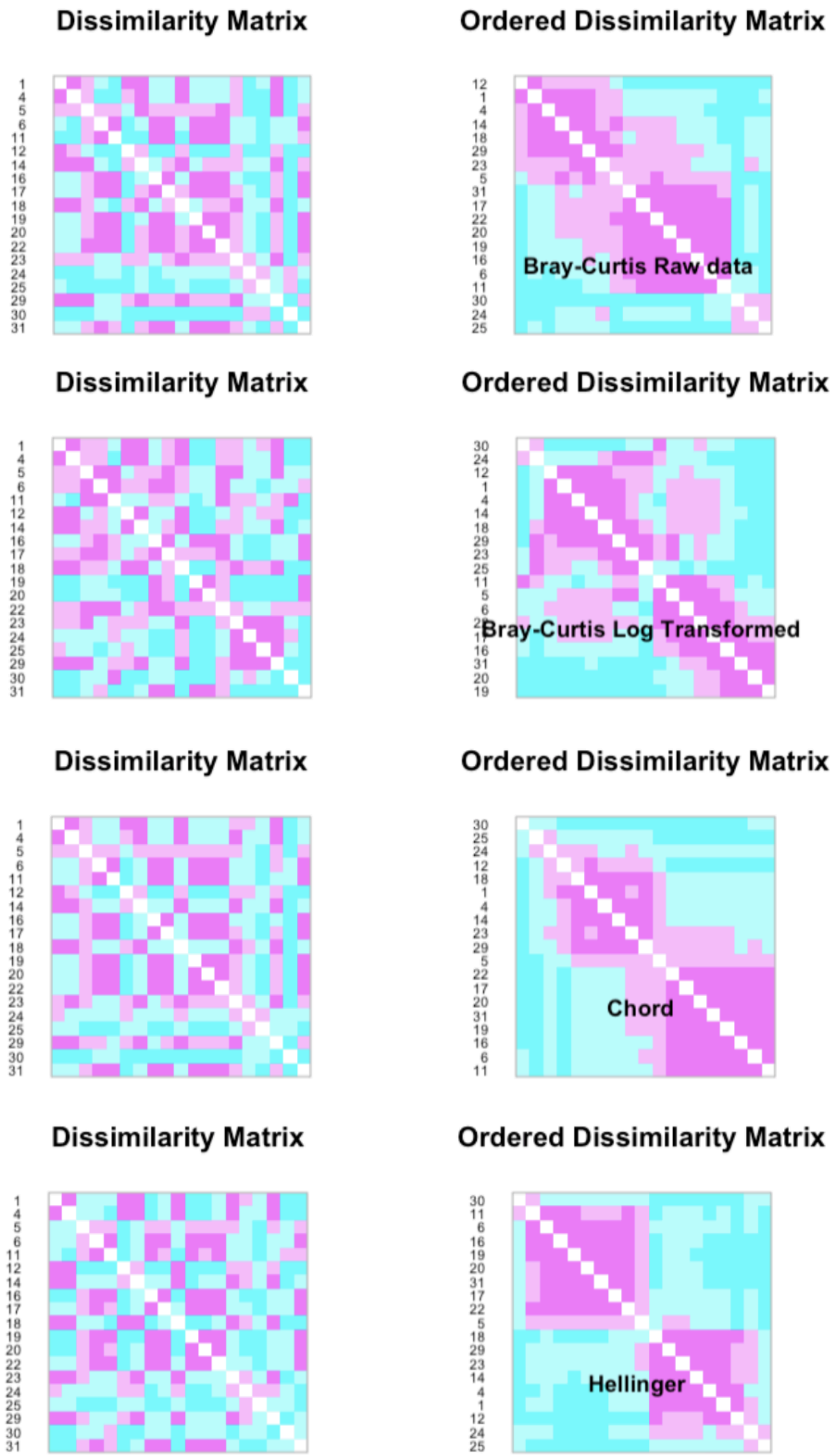
Figure 16: PCA of the biodiversity index with sites representation.

We observe a correlation between all biodiversity indices except NO, which represents species richness. This observation was expected, as J (Pielou evenness), E1 (Shannon evenness), and E2 (Simpson evenness) are calculated based on NO, which serves as their denominator. Sites with the highest biodiversity appear to be sites 19, 20, 22, 29, 24, and 5, which is consistent with what was previously discussed in question 6.

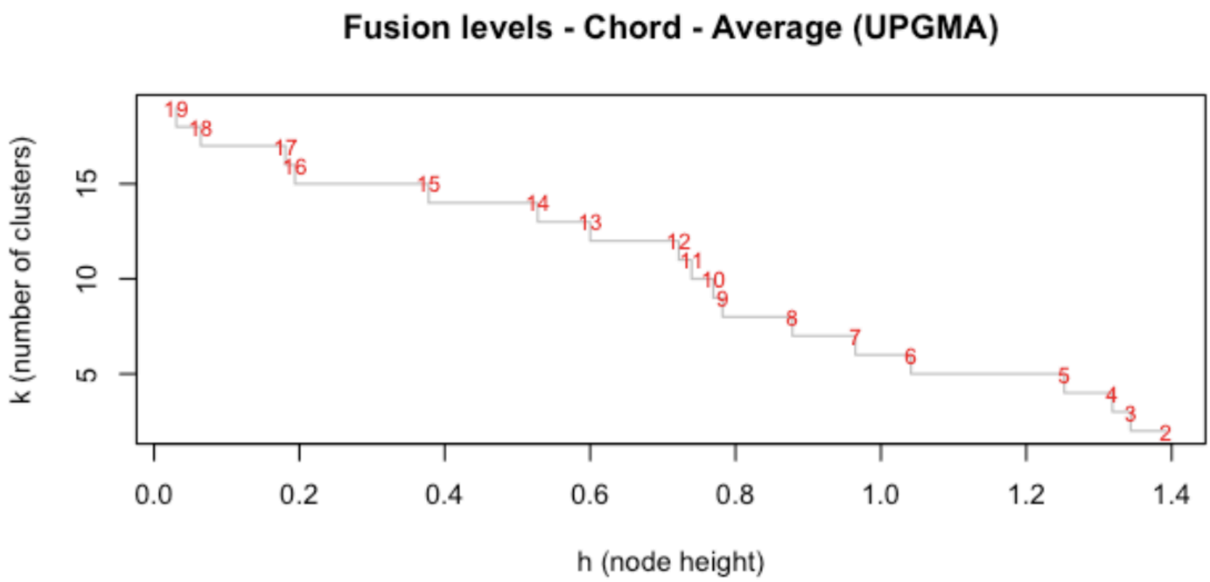
Appendix 1.A : Distance matrices between sites and species diversity.



Appendix 1.B : Distance matrices between sites and environmental variables.



Appendix 2: Fusion level with Chord average for species analysis



Appendix 3: Fusion level with Chord average for environmental analysis.